
Predictive Evaluation Competition

Daphne Barretto
daphnegb@stanford.edu

Izhan Hamza
izhamza@stanford.edu

Shannon Komguem
skomguem@stanford.edu

Abstract

We address the Predictive AI Evaluation Challenge, a cold-start problem in which a model must predict $\hat{p} = P(\text{correct} \mid \text{subject, item})$ given only four text fields and no prior observations of the test items. Our pipeline combines a 2PL Item Response Theory backbone with a sentence-transformer encoder (`all-mpnet-base-v2`)—shared across subject, item, benchmark, and condition—feeding a 5-layer MLP head; logits then pass through post-hoc Platt and temperature scaling, and the $K=5$ adaptive labels are folded in as a single per-subject mean-residual offset. Our first leaderboard submission *regressed* to -1.01 NLL relative to a -0.70 in-team baseline; three diagnostic submissions and a single scalar temperature ($T \approx 4.07$, fit on a leave-two-benchmarks-out hold-out) recover the curve to -0.65 , and an equal-weight arithmetic blend with the per-subject training-time mean accuracy ($0.5 p_{T\text{-scaled}} + 0.5 \bar{y}_s$) reaches -0.61 — beating the in-team baseline by 0.09 NLL with AUC stable at $0.62\text{--}0.63$ throughout. A six-variant Part V coverage sweep further bounds the maximum contribution of *all* calibration, content, and metadata machinery at ≈ 0.04 NLL above the $-\ln 2$ floor. We argue that for cold-start $P(\text{correct})$ prediction the dominant lever is controlled under-confidence at test time, not encoder choice.¹

1 Problem Formulation & Method Description

1.1 Problem Setting

Let $\mathcal{S}$ be subjects (language-model checkpoints) and $\mathcal{I}$ items (benchmark questions). Training data from `aims-foundations/measurement-db` forms a sparse response matrix $R \in \{0, 1, \text{NaN}\}^{|\mathcal{S}| \times |\mathcal{I}|}$ where $R_{ij} = 1$ iff subject i answers item j correctly under condition c . At test time the scorer asks for $\hat{p} \in (0, 1)$ on items whose benchmarks do *not* appear in training, scored by mean Bernoulli log-likelihood (higher is better; $\hat{p}=0.5$ gives $-\log 2 \approx -0.693$). The platform optionally reveals $K=5$ ground-truth labels per round (chosen by our `acquisition_function`) *before* the prediction round on the remainder—the only direct observation of the held-out distribution available at inference.

1.2 Method Overview

The pipeline factorizes into a low-rank *ability backbone* and a content-aware *residual head*, in the spirit of factor-augmented psychometric regression. (i) A 2PL IRT model fit on the dense submatrix (≥ 5 observed responses; $\sim 33\text{K}$ items) yields a subject ability θ_i and per-item (a_j, b_j) . (ii) A shared `all-mpnet-base-v2` encoder embeds all four text fields into 768-d vectors, so that benchmarks and conditions unseen at training time still receive meaningful representations rather than the all-zero categorical indicator a one-hot scheme would produce. (iii) A 5-layer MLP head maps the 3075-d concatenation $[\theta, \mathbf{e}_{\text{subj}}, \mathbf{e}_{\text{item}}, \mathbf{e}_{\text{bm}}, \mathbf{e}_{\text{cond}}, \mu_{\text{subj}}, \mu_{\text{bm}}]$ to a logit (architecture in Appendix B.1; $\mu_{\text{subj}}, \mu_{\text{bm}}$ are empirical pass-rates). (iv) Post-hoc Platt re-scaling $\sigma(\alpha\ell + \beta)$ fit on a 10% in-distribution

¹Code: <https://github.com/daphne-barretto/prediction-evaluation-challenge>.

hold-out, then temperature scaling $\sigma(\ell/T)$? fit on a cold-start hold-out. (v) At inference the $K=5$ labels are folded in as a single per-subject mean-residual offset $\delta_s = \frac{1}{K} \sum_k (\hat{p}_k - y_k)$, subtracted from subsequent predictions; outputs are clipped to $[10^{-3}, 1-10^{-3}]$.

Adaptive labeling. We pick the K items that maximize $\text{score}(x) = (1 - |2\hat{p}(x) - 1|) \cdot \frac{1}{1+n_c(x)}$, the product of model uncertainty and a diversity penalty over 64 pre-fit k -means centroids on item embeddings. Either factor alone is degenerate (uncertainty collapses semantically; diversity wastes labels on easy items); the product favors items that are both informative and representative of distinct regions of the input distribution. Revealed labels feed the offset above. We leave θ unchanged at test time because $K=5$ Newton–Raphson updates were empirically unstable (Appendix A.2).

Training procedure. We fit on $\sim 2.7\text{M}$ response rows held out from `bfc1` and `matharena`; total wall-clock time is $\approx 20\text{--}25$ min on a single T4 GPU. Full pipeline (compute, data binarization, artifact list) in Appendix B.2.

2 Training Dataset

We use the public `aims-foundations/measurement-db` parquet shards ? (no additional curated data). After join and binarization, the usable training set contains $\sim 3.4\text{M}$ response rows over 909 subjects, $\sim 104\text{K}$ items, 15 benchmarks (13 train, 2 cold-start hold-out: `bfc1` and `matharena`), and 48 conditions; overall pass-rate ~ 0.65 . Of those items, $\sim 33\text{K}$ have ≥ 5 recorded responses and participate in the dense IRT fit; the remainder receive the mean (a, b) as fallback. Downstream-split row counts are in Table 6 (Appendix B.3); detailed preprocessing, label binarization, and cold-start split construction are also in Appendix B.3.

3 Experimental Results & Ablation Studies

3.1 Validation Setup

We use two complementary validation regimes plus the leaderboard. The *in-distribution Platt hold-out* is a 10% uniform random hold-out of non-cold-start rows on which Platt (α, β) are fit; we also report its post-Platt NLL as an in-distribution calibration sanity check. The *cold-start hold-out* removes a fixed pair of benchmarks (`bfc1`, `matharena`) entirely from training to estimate the OOD gap, since their items, conditions, and item families are unseen at training time. We additionally report leaderboard NLL on Codabench.

3.2 Main Results

Table 1 reports NLL across the three regimes plus four diagnostic Codabench submissions that progressively disable test-time components and then add post-hoc temperature scaling ($T=4.073$ fit on the cold-start hold-out).

Headline finding. Sub 1 fell to -1.01 , *worse* than constant $\hat{p}=0.5$. Removing Platt (sub 2) and the per-subject offset (sub 3) recover 0.16 NLL; post-hoc T-scaling (sub 5) adds a further 0.20, landing at -0.65 . This beats the in-team baseline (-0.70) by 0.05 NLL with only 0.07 NLL separating it from the public top entry (-0.58). AUC is stable at 0.62–0.63 across all four submissions — the entire 0.36-NLL improvement comes from *calibration*, not from any change to the ranking signal. The temperature was fit on the cold-start hold-out (-0.665) and generalised almost perfectly to the leaderboard (-0.650); a 0.015 NLL gap.

Equivalence of three calibration families. We compared three one- and two-parameter calibrators on the same cold-start hold-out ($N \approx 4 \times 10^5$ logits): pure T-scaling, prior shrinkage $\alpha\sigma(\ell) + (1-\alpha)/2$, and their composition. All three converge to within ~ 0.001 NLL ($T^*=4.07 \Rightarrow -0.6648$; $\alpha^*=0.386 \Rightarrow -0.6643$; joint -0.6639). Full T -vs- α contour in Appendix A.1.

Table 1: NLL and ROC AUC by evaluation regime and ablation submission. “offset” is the $K=5$ per-subject mean-residual; “T-scaling” applies $\sigma(\text{logit}/T)$ with $T=4.073$. Reference is $-\log 2 \approx -0.693$.

Regime / Submission	NLL	AUC
Constant prediction $\hat{p} = 0.5$	−0.693	0.500
In-distribution Platt hold-out (Platt)	−0.370	—
Cold-start hold-out, {bfcl,matharena} (pre-Platt)	−0.832	—
Cold-start hold-out, {bfcl,matharena} (Platt)	−0.884	—
Cold-start hold-out, {bfcl,matharena} (T-scaling, $T=4.073$)	−0.665	—
Codabench: Platt + offset (sub 1, round 742341)	−1.012	0.620
Codabench: identity-Platt + offset (sub 2)	−0.930	0.630
Codabench: identity-Platt, no offset (sub 3)	−0.850	0.630
Codabench: identity-Platt, no offset, T-scaling (sub 5)	−0.650	0.630
Previous in-team baseline (40976e8, leaderboard)	−0.700	—
Top of public leaderboard (reference)	−0.580	—

3.3 Ablation Studies

We compare against an earlier in-team baseline (commit 40976e8 on the izhan branch) which shipped a MiniLM-L3 encoder, one-hot benchmark/condition features, and a Newton–Raphson θ update; it scored −0.70 on the leaderboard. The component-by-component delta (Table 3, Appendix A.2) shows that the encoder, feature scheme, MLP shape, calibration, and K -shot use all changed simultaneously. The submission climb of §3.2 then attributes the recovery to the calibration column of that table: removing Platt, removing the per-subject offset, and finally adding T-scaling are sufficient to beat the baseline by 0.05 NLL on the leaderboard, without touching the encoder upgrade or the text-embedded categorical scheme.

3.4 Strategy coverage: the calibration climb and family map

To bound how much of the leaderboard NLL is recoverable from calibration alone, we ran a deliberate set of ablations spanning the two scaffolds in the Discussion 6 tutorial: a seven-rung calibration climb (R1: $\hat{p}=0.5$ to R7: full PGE with $K=5$ adaptive labels) and a six-family option space (**F1** priors, **F2** IRT alone, **F3** content, **F4** metadata, **F5** local LLM judge, **F6** adaptive). Table 2 maps each submission onto these axes; per-submission leaderboard scores are in Table 4 (Appendix A.3).

Table 2: Coverage of Discussion 6 strategy axes. Rows are the seven-rung calibration climb; columns are the six families. ✓ marks ablations we shipped; × marks families we deliberately did not pursue (F2 IRT-only fails cold-start by design ?; F5 local LLM judge does not fit the per-tier sandbox compute budget).

	F1 priors	F2 IRT-only	F3 content	F4 metadata	F5 local LLM	F6 adaptive
R1: $\hat{p} = 0.5$	✓					
R2: global mean	✓					
R3: smoothed subject mean	✓			✓		
R4: per-(bench, cond) mean	✓			✓		
R5: + content (text → params)			✓		×	
R6: + IRT θ (full PGE)		×	✓			
R7: + adaptive $K=5$ labels			✓			✓

Findings (full results in Appendix A.3). The four prior submissions (R1–R4) bound the maximum contribution of *all* calibration+content+metadata machinery to ≈ 0.04 NLL above the $-\ln 2$ floor (sub 9 scores −0.69, matching the theoretical $-\ln 2$). The robustness sweep (subs 5–7) confirms the calibration manifold is broad in T : all three score −0.65. Sub 8 (per-subject offset) regressed by

0.10 NLL, motivating sub 14’s single-global-shift design (a no-op at -0.65). **Sub 13** ($0.5 p_{\text{sub } 5} + 0.5 \bar{y}_s$) **lands at -0.61 , a new best by 0.04 NLL over sub 5.**

3.5 Leaderboard Performance

We submitted 18 models. Sub 1 (mpnet_platt_meanresid) returned -1.012 , a 0.31 NLL regression versus the in-team baseline (dissected in §4.1). Subs 2, 3, 5 recover the curve to -0.65 via diagnostic component disabling plus T-scaling. **Sub 13** ($0.5 p_{\text{sub } 5} + 0.5 \bar{y}_s$) then jumps to -0.61 , our new best; the remaining variants (subs 6–20, §3.4) extend the family map (Appendix A.3).

4 Analysis of Failure Modes & Interesting Patterns

4.1 Error Analysis

Ranking versus calibration. An AUC of 0.62 on the Codabench test set means sub 1 still orders pairs of (correct, incorrect) items correctly 62% of the time — well above chance. The catastrophic -1.01 NLL is therefore a *confidence* failure, not a discrimination failure: many of the model’s most confident predictions are wrong, which costs heavily under log-loss while costing little under AUC. We attribute the collapse to three compounding mechanisms (detailed in Appendix A.4): Platt sharpening $\alpha \approx 1.15$ pushes confidently-wrong OOD predictions toward $\{0, 1\}$; the encoder upgrade trades implicit-regularisation capacity for tighter in-distribution fit; and the per-subject mean-residual offset is exquisitely sensitive to which 5 items the platform happens to reveal.

4.2 Interesting Patterns

Distribution shift dominates the error budget. Going from a 10% random hold-out (NLL -0.37) to a leave-two-benchmarks-out hold-out (-0.88) to the Codabench test set (-1.01 for sub 1) shows a clean monotonic relationship between distribution shift and NLL — a gap of ≈ 0.6 NLL that cross-validation on the training distribution would have completely missed.

Calibration, not representation, is the binding lever. The single change that recovers the leaderboard score is post-hoc temperature scaling: parameter-free at training time, costs ~ 10 s to fit on the cold-start hold-out, and lands at -0.65 (sub 5). The encoder upgrade and the text-embedded categoricals are kept across all recovered submissions, yet contribute essentially nothing to the climb $3 \rightarrow 5$ — the entire 0.20 NLL recovery comes from a single scalar $T \approx 4$. The empirical equivalence of T-scaling, prior shrinkage, and their composition (§3.2) extends this: *any* 1- or 2-parameter post-hoc calibrator wins as long as it shrinks predictions toward the prior on the cold-start hold-out.

4.3 Limitations

- *Cold-start gap is large.* Even with text-embedded benchmarks, our pipeline degrades from -0.37 (in-distribution) to -0.83 (cold-start) to -1.01 (Codabench) before calibration — the MLP has not learned a truly transferable map from item text to correctness.
- *No encoder fine-tuning.* We use mpnet zero-shot; fine-tuning on $(\text{item}, \text{label})$ pairs is the natural next step but exceeded the compute budget.
- *$K=5$ is a hard signal-to-noise floor.* A single global scalar offset (sub 14) is the most we trust ourselves to extract from 5 labels per category; any per-subject or per-cluster split is empirically unstable (sub 8 regressed by 0.10 NLL).

5 Conclusion

We treated the Predictive Evaluation Competition as a calibration problem dressed up as a representation-learning problem. Sub 13’s blend with the per-subject training-time mean (-0.61) is our strongest result; three one- and two-parameter post-hoc calibrators converge to within 0.001 NLL (§3.2); any $T \in [3, 5.5]$ yields the same -0.65 leaderboard score (subs 5–7); and the Part V coverage map (§3.4) bounds the maximum contribution of *all* calibration, content, and metadata machinery at ≈ 0.04 NLL above the $-\ln 2$ floor. The remaining 0.03 NLL gap to the top public entry must therefore come from a better content head or from a learned use of the adaptive-label channel.

A Additional Results

A.1 Full T -vs- α calibration map

We dumped $N \approx 4 \times 10^5$ raw MLP logits from the trained sub 5 pipeline on the leave-two-benchmarks-out hold-out (bfc1+matharena) and swept three one- and two-parameter post-hoc calibrators by exact NLL minimization:

- **Temperature scaling** $\hat{p} = \sigma(\ell/T)$, $T \in [0.5, 10]$. The optimum is $T^* = 4.073$ with $\text{NLL} = -0.6648$.
- **Prior shrinkage** $\hat{p} = \alpha \sigma(\ell) + (1 - \alpha)/2$, $\alpha \in [0, 1]$. The optimum is $\alpha^* = 0.386$ with $\text{NLL} = -0.6643$.
- **Composition** $\hat{p} = \alpha \sigma(\ell/T) + (1 - \alpha)/2$ jointly in (T, α) . The optimum is $(T, \alpha) \approx (2.1, 0.6)$ with $\text{NLL} = -0.6639$.

All three converge to within 0.001 NLL of each other, and the (T, α) loss surface is essentially a single ridge: any pair on the line $T \cdot (2\alpha/(1 + \alpha)) \approx 1.6$ gives near-optimal NLL. The leaderboard sweep (subs 5–7, $T \in \{3.0, 4.07, 5.5\}$, all scoring -0.65) sits squarely on this ridge.

A.2 Component ablation versus the prior in-team baseline

Table 3 compares the prior in-team baseline (commit 40976e8 on the izhan branch, leaderboard -0.70) against the sub 5 pipeline. Five components changed simultaneously; the diagnostic submission climb of §3.2 attributes the $+0.05$ NLL recovery to the calibration column alone.

Table 3: Component-by-component delta, baseline (40976e8) vs. sub 5.

Component	Baseline	Sub 5
Text encoder	paraphrase-MiniLM-L3-v2 (17M, 384-d)	all-mpnet-base-v2 (110M, 768-d)
Benchmark/condition	one-hot	text-embedded (shared encoder)
MLP head	4-layer 256-128-64-1	5-layer 512-256-128-64-1, LN+dropout
Calibration	none	Platt (identity in sub 5) + T-scaling ($T=4.07$)
$K=5$ use	Newton–Raphson on θ	per-subject mean residual (disabled in sub 5)
Leaderboard NLL	-0.70	-0.65

Why each change was expected to help.

- *Encoder upgrade.* all-mpnet-base-v2 is a more accurate sentence embedder on benchmark NLI tasks; we expected it to give better semantic separation between content categories.
- *Text-embedded categoricals.* One-hot indicators are all-zero for benchmarks unseen at training time; text embeddings give a non-degenerate representation that should at least transfer zero-shot.
- *Wider MLP.* 110M-d encoder outputs deserve more head capacity than a 4-layer 256-width path; LayerNorm + dropout were added to control the corresponding overfitting risk.
- *Platt re-calibration.* A logistic re-scaler on raw logits is the textbook fix for over- or under-confident classifiers.
- *Mean-residual offset.* Replaces the prior Newton–Raphson θ update with something that does not assume the 5-shot labels are unbiased samples from the subject’s overall accuracy.
- *T-scaling (post-hoc, sub 5).* A single scalar ($T=4.07$) fit on a leave-two-benchmarks-out hold-out; this is the change that actually moved the leaderboard.

A.3 Full Part V leaderboard sweep

Table 4 reports leaderboard NLL for each Part-V variant. The seven-rung calibration climb (R1–R7) is read from the slide-29 scaffold; subs 6, 7, 8 are robustness sweeps over T and the offset; subs 13–16 are the family-coverage variants.

Table 4: Part V ablation grid. “LB NLL” is the Codabench leaderboard score; “–” indicates the submission was still pending in the Codabench queue at report compilation time.

Sub	Variant	Rung / Family	LB NLL
9	$\hat{p} \equiv 0.5$	R1 (F1 priors)	−0.69
10	global pass-rate 0.645	R2 (F1 priors)	–
15	smoothed subject mean	R3 (F1+F4)	–
16	per-benchmark mean	R4 (F1+F4)	–
5	T-scaled mpnet, $T=4.07$	R6/R5 (F3 content)	−0.65
6	T-scaled mpnet, $T=3.0$	R6 robustness	−0.65
7	T-scaled mpnet, $T=5.5$	R6 robustness	−0.65
8	sub 5 + per-subject offset	R7 (F3+F6)	−0.75
13	$0.5 p_{\text{sub } 5} + 0.5 \bar{y}_s$	R7 (F3+F4 blend)	− 0.61
14	sub 5 + shrunk global mean-residual	R7 (F3+F6 light)	−0.65
17	$0.3 p_{\text{sub } 5} + 0.7 \bar{y}_s$	blend α sweep	–
18	$0.7 p_{\text{sub } 5} + 0.3 \bar{y}_s$	blend α sweep	–
19	$0.5 p_{\text{sub } 5} + 0.5 \mu_{\text{global}}$	prior-pull control	–
20	sub 13 + shrunk global mean-residual	blend + K=5 shift	–

R1–R4 (subs 9, 10, 15, 16). Sub 9 ($\hat{p} \equiv 0.5$) scored −0.69, matching $-\ln 2 \approx -0.693$ to within rounding. The four trivial floors bound the contribution of all calibration, content, and metadata machinery in our system: if sub 13 (−0.61) only beats the R1 constant by 0.08 NLL, then no combination of further calibration, prior shrinkage, or per-(subject, benchmark) lookup can push the floor much further; the residual gap to the public top entry (−0.58) must come from better content representation or a learned use of the adaptive labels.

T-robustness (subs 5, 6, 7). All three scored −0.65 on the leaderboard. This confirms the broad-ridge result from Appendix A.1: the leaderboard distribution is within the calibration manifold of our cold-start hold-out, and sub-percent-NLL gains are not achievable by fine-tuning T .

Sub 8 post-mortem. Re-enabling the per-subject mean-residual offset on top of T-scaling lost 0.10 NLL. With only $K=5$ labels per category the offset is too noisy: a single subject whose 5 revealed items happen to be hard pushes *all* of that subject’s predictions far from 0.5, where they incur disproportionate log-loss. Sub 14 was designed as a conservative response: aggregate all $K \times$ categories labels into a single global shift, shrink it by $\lambda=25$ ridge, and cap magnitude at $\pm 5\text{pp}$. Sub 14 scored −0.65 on the leaderboard, a no-op on top of sub 5: with T-scaled predictions already sitting near 0.5, the mean residual is dominated by noise.

Sub 13. The blend $0.5 p_{\text{sub } 5} + 0.5 \bar{y}_s$ scored −0.61, our new best by 0.04 NLL. The mechanism is variance reduction: T-scaled mpnet outputs sit near 0.5 (high entropy, low ranking signal) and the per-subject training-time mean $\bar{y}_s$ is a zero-variance estimate of overall accuracy on the items the subject has already answered. Equal-weight arithmetic averaging preserves most of sub 5’s ranking while pulling each prediction toward a per-subject prior, which is exactly the configuration that under-confident, well-ranked predictors benefit from.

Sub 13 ablation sweep (subs 17–20). To attribute sub 13’s gain we built four follow-ups: subs 17 ($\alpha=0.3$) and sub 18 ($\alpha=0.7$) sweep the blend weight to locate α^* (sub 13 is $\alpha=0.5$); sub 19 replaces $\bar{y}_s$ with the single global mean ($\mu_{\text{global}}=0.645$) as a prior-pull control (if sub 19 ties sub 13, per-subject identity is irrelevant; if it loses, the per-subject signal is doing real work); sub 20 stacks sub 14’s shrunk-capped global K-shift onto sub 13, since the blend’s predictions sit near 0.57 rather than 0.5 and may now expose useful mean-residual signal that sub 14 could not. All four were in the queue at compilation time.

Strategies not pursued. F2 (IRT-only) fails cold-start by design: with no items in common with training, the difficulty parameters (a_j, b_j) have no support and reduce to a global prior $?$. F5 (local LLM judge) does not fit the Codabench $?$ per-tier sandbox compute budget. Both are marked $\times$ in Table 2.

A.4 Error analysis: three compounding mechanisms

(i) **Platt sharpening amplifies OOD errors.** Our fitted Platt slope $\alpha \approx 1.15$ is greater than 1, meaning Platt *sharpen*s the predicted probabilities slightly. This reduces in-distribution NLL but pushes any confidently-wrong prediction further toward the $\{0, 1\}$ boundary. We had already observed this on our own cold-start hold-out (raw $-0.83 \rightarrow$ Platt -0.88); the leaderboard appears to lie further OOD than our hold-out, so the same effect is presumably larger there. Sub 2 (identity Platt) recovered 0.08 NLL.

(ii) **Capacity-induced overfitting.** The encoder upgrade took us from 17M to 110M parameters and replaced sparse one-hots with 768-d benchmark and condition embeddings. With the same training data and the same MLP head shape, this gives the model many more degrees of freedom for fitting training-distribution-specific patterns. The baseline’s lower capacity may have been acting as implicit regularization toward more transferable solutions.

(iii) **Subject offset is noisy in 5-shot.** The mean-residual offset shifts *all* of a subject’s predictions by a single constant computed from $K=5$ labels. If those five labels happen to land on items where the model is catastrophically wrong, the constant moves already-good predictions in the wrong direction. The baseline’s Newton–Raphson update adjusted θ rather than the output, which was, in principle, more robust to such mismatches. Sub 3 (offset disabled) recovered another 0.08 NLL.

B Implementation Details

B.1 Full model architecture

The 2PL IRT backbone $P(R_{ij}=1 \mid \theta_i, a_j, b_j) = \sigma(a_j(\theta_i - b_j))$ is fit on the dense submatrix (≥ 5 observed responses per item; $\sim 33K$ items) by NaN-masked BCEWithLogitsLoss; 1000 epochs of Adam ($\eta = 10^{-2}$, $\text{wd} = 0$) on a T4 GPU. The shared encoder is `sentence-transformers/all-mpnet-base-v2` (110M params, 768-d output), run in eval mode with mean-pooling. The 3075-d input $[\theta, \mathbf{e}_{\text{subj}}, \mathbf{e}_{\text{item}}, \mathbf{e}_{\text{bm}}, \mathbf{e}_{\text{cond}}, \mu_{\text{subj}}, \mu_{\text{bm}}]$ is consumed by the MLP in Table 5; training uses Adam ($\eta = 3 \times 10^{-3}$, $\text{wd} = 10^{-4}$, batch 2048), 30 epochs, NaN-masked BCE loss, ReduceLROnPlateau on the in-distribution Platt hold-out, early stop on the cold-start hold-out.

Table 5: 5-layer MLP head used by sub 5.

Layer	Shape	Notes
Linear 1	3075 $\rightarrow$ 512	LayerNorm, ReLU, Dropout 0.2
Linear 2	512 $\rightarrow$ 256	LayerNorm, ReLU, Dropout 0.2
Linear 3	256 $\rightarrow$ 128	LayerNorm, ReLU, Dropout 0.1
Linear 4	128 $\rightarrow$ 64	LayerNorm, ReLU
Linear 5	64 $\rightarrow$ 1	raw logit

Calibration. Platt fits a 1-d logistic regression $\sigma(\alpha\ell + \beta)$ on the 10% in-distribution hold-out by L-BFGS. Temperature is fit on the cold-start hold-out by `scipy.optimize.minimize_scalar` over $T \in [0.5, 10]$. Final inference is $\hat{p} = \sigma((\alpha\ell + \beta)/T) - \delta_s$, clipped to $[10^{-3}, 1 - 10^{-3}]$.

B.2 Training pipeline, compute, and artifacts

Training runs on a personal Modal workspace (ap-2KtFZJ7o0hi57ZvqwvsMTc), single NVIDIA T4 GPU, 64 GB RAM. Wall-clock ≈ 20 –25 min per run. Labels are binarized via $y_{ij} = \mathbb{I}[\text{score} \geq 0.5]$. The cold-start split holds `bfc1` and `matharena` out of *all* training (IRT, MLP, Platt); the Platt hold-out is a 10% uniform random hold-out of the remaining rows. Memory engineering: the dense submatrix is streamed shard-by-shard from parquet to avoid materializing the full $909 \times 104K$ matrix in RAM; embeddings are cached to disk per text field. Artifacts shipped in the submission ZIP: `mlp.pt`, `platt.json`, `temperature.json`, `theta_lookup.json`, `subject_emb_index.json`, `subject_embeddings.npy`, `benchmark_emb_index.json`, `benchmark_embeddings.npy`,

condition_emb_index.json, condition_embeddings.npy, subject_mean_acc.json, benchmark_mean_acc.json, global_mean_acc.json, centroids.npy.

B.3 Dataset preprocessing

Table 6: Training-data composition and downstream splits.

Quantity	Count
Subjects	909
Unique items	103,983
Benchmarks (train / cold-start hold-out)	13 / 2
Cold-start held-out benchmarks	bfcl, matharena
Cold-start hold-out rows	397,484
Platt hold-out rows (10%)	303,543
MLP training rows	2,731,896

Subject text. Concatenation of the subject’s HuggingFace identifier, organization, and any free-text description, normalized by lowercasing and stripping `<eos>`/`<pad>` tokens.

Condition normalization. Conditions are stripped of trailing whitespace and collapsed to lower-case; the resulting 48 canonical strings are embedded by the shared encoder.

Filtering. Rows with NaN labels are dropped before binarization; subjects with < 5 total responses are kept but receive the global ability prior $\theta = 0$.

Feature standardization. μ_{subj} and μ_{bm} are computed on the MLP training rows only (i.e. excluding both the cold-start and Platt hold-outs) to prevent leakage; both are clipped to $[10^{-3}, 1 - 10^{-3}]$ before consumption.

C AI Disclosure

C.1 Plagiarism, Bias, and Inaccuracies Statement

We used only the public Hugging Face dataset `aims-foundations/measurement-db`, with no additional curated or scraped data. We did not redistribute its contents; all derived artifacts (embeddings, mean-accuracy tables, MLP weights) were computed in-house from the published parquet shards. We are aware that the response matrix is biased toward English-language items and toward subjects from a handful of large U.S./European labs, and we treat the resulting predictions accordingly: any inference about the performance of an underrepresented subject on an underrepresented benchmark is unreliable, and our cold-start hold-out (bfcl + matharena) was chosen to expose, not hide, this limitation. We made no claims of correctness about specific items: a $\hat{p}$ value is a calibrated probability, not a verification of the underlying item–subject pair. Code and artifacts are available at the repository linked from the title page.

C.2 AI Tools Reflection

We used the GitHub Copilot CLI as a paired engineering and writing assistant throughout this project. Concretely, AI assistance helped us (a) extract and summarize the Discussion 6 tutorial slides so that we could plan the Part V ablation grid; (b) translate the textual calibration recipes into runnable Python variants for the `predict()` interface; (c) draft this report’s LaTeX and run syntactic checks. Each AI-generated edit was reviewed and either rewritten or accepted by a human author, and the underlying model artifacts (mpnet embeddings, MLP weights, IRT θ , temperature scalar) were trained and verified by us rather than produced by a language model. Going forward, we expect to keep using AI for the same role—fast pattern-matching against published technique catalogs—while reserving the experimental design, error analysis, and writing of insight to ourselves.

C.3 Impact Statement

Predictive evaluators like the one we built lower the cost of deciding which AI subject to deploy on which task, which is double-edged: they speed up beneficial deployments but can also encourage skipping direct measurement when a predicted probability “looks good enough.” We mitigate this in three ways. First, our final model deliberately publishes *flat* confidence ($T^* \approx 4$ effectively narrows every sigmoid output toward 0.5), so a user reading our predictions cannot mistake them for sharp claims about correctness. Second, the cold-start hold-out in §3.2 establishes the achievable NLL ceiling (≈ -0.66); any downstream user who needs sharper confidence should run the actual evaluation instead of trusting the predictor. Third, our code adheres to the competition’s network-isolation rule: no test-time data leaves the sandbox, no external APIs are queried, and no subject- or item-level data persists across rounds.